

RADBOUD UNIVERSITY



Faculty of Science

Optimizing Local Movement Rules for Heterogeneous Crowd Evacuation with a Genetic Algorithm

Natural Computing: NWI-IMC042

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1 Problem Statement

Efficient evacuation is an important problem in settings where many people need to move through a limited space under time pressure. In such situations, global crowd behaviour emerges from many local interactions, such as moving toward an exit, avoiding nearby people, and responding to congestion near bottlenecks. Because of this, evacuation is a useful example of collective behaviour in which simple individual rules can lead to complex group-level outcomes such as clogging, uneven flow, and longer evacuation times.

This project focuses on a simplified hospital-inspired evacuation scenario. In hospital-like environments, the crowd is often heterogeneous rather than homogeneous: some individuals can move quickly and independently, while others have reduced mobility and may be more strongly affected by congestion. We model this heterogeneity by distinguishing between *high-mobility* and *low-mobility* agents. This makes the problem scientifically interesting, because a rule set that improves average evacuation performance may not benefit all groups equally. In particular, a strategy that mainly helps fast-moving agents could still perform poorly for slower and more vulnerable individuals.

Existing work has already studied pedestrian evacuation, heterogeneous populations, and optimization of simulation parameters. Evolutionary and related methods have been used, for example, to calibrate pedestrian models or optimize evacuation-related settings. For that reason, our project does not claim to introduce a completely new problem. Instead, we build on this line of work by studying a simple and controlled setting in which a genetic algorithm is used to tune local movement parameters in a heterogeneous crowd model. Our extension is to explicitly evaluate whether such tuned parameters improve evacuation performance in mixed-mobility crowds, and whether they remain effective across different scenarios.

The main goal of this project is to investigate whether a genetic algorithm can discover local movement parameters that improve evacuation performance in a heterogeneous crowd model with high- and low-mobility agents.

Our main research question is:

Can a genetic algorithm discover local movement parameters that improve evacuation performance in a heterogeneous crowd model with high- and low-mobility agents?

To answer this question, we focus on three sub-questions:

1. Can a genetic algorithm find parameter settings that outperform random search and fixed heuristic settings on evacuation performance?
2. Do the evolved parameters improve outcomes for both mobility groups, or do they mainly benefit the high-mobility agents?
3. Do the evolved parameters generalize to different crowd densities, mobility ratios, or room layouts?

To study this, we will use a simplified 2D agent-based evacuation model. Agents will move toward an exit while following a small number of local rules, such as attraction toward the exit, avoidance of nearby agents, and possibly a weak tendency to follow local movement direction. The model will contain both high- and low-mobility agents, where the two groups mainly differ in preferred movement speed. A genetic algorithm will then optimize a small set of parameters controlling these local movement rules.

Each candidate parameter setting will be evaluated by running the simulation and measuring outcomes such as total evacuation time, congestion near the exit, collisions or near-collisions, and evacuation time per mobility group. We will compare the genetic algorithm against random search and fixed heuristic parameter settings under the same evaluation setup. This approach is suitable because the relationship between local

movement parameters and global evacuation behaviour is non-linear and difficult to tune by hand, while genetic algorithms are well suited for black-box optimization problems in which the quality of a solution can be measured through simulation.

In summary, this project studies whether evolutionary optimization can improve local movement rules for heterogeneous crowd evacuation in a simple hospital-inspired setting. By focusing on mixed-mobility crowds, group-specific outcomes, and generalization across scenarios, the project aims to provide a clear and feasible contribution that combines collective behaviour modelling with evolutionary algorithms.

2 Related Work

Helbing, D., & Molnár, P. (1995). "Social force model for pedestrian dynamics." [4]

Helbing and Molnár (1995) introduced the social force model, one of the best-known models for simulating pedestrian movement. In their work, pedestrians are influenced by local forces such as attraction toward a goal and repulsion from other pedestrians and walls. This model is important for our project because it shows how simple local interaction rules can lead to complex crowd-level behaviour such as self-organization, congestion, and bottlenecks. We build on this same general idea by using a simplified local-rule crowd model in which agents move toward an exit while interacting with nearby agents. However, rather than only studying the resulting crowd behaviour, we use a genetic algorithm to search for parameter settings that improve evacuation performance in a heterogeneous crowd.

Cui, G., Yanagisawa, D., & Katsuhiko, N. (2022). "A Data Driven Approach to Simulate Pedestrian Competitiveness Using the Social Force Model." [2]

Cui et al. (2022) applied a differential evolutionary algorithm to calibrate the continuous local movements parameters of a social force model to simulate pedestrian dynamics. The algorithm searched for different combinations of parameters, such as desired speed or avoidance strength. The goal was to accurately reproduce the trajectory patterns of "patient" versus "impatient" pedestrians that move through a narrow bottleneck. It relates directly to our project because we will use an EA to optimize local movement parameters in an agent-based pedestrian simulation. We also will encode behavioural differences as parameter sets. However, Cui et al. used the EA to calibrate the algorithm to match the empirical evidence. They did not use it to test whether evolutionary search can be used to improve global evacuation metrics. We will also use an EA to a pedestrian model, but we will use it to search for a set of rules that minimizes evacuation time and congestion.

Cui, G., Yanagisawa, D., & Nishinari, K. (2021). "Incorporating genetic algorithm to optimise initial condition of pedestrian evacuation based on agent aggressiveness." [3]

Cui et al. (2021) incorporated a genetic algorithm into a floor field cellular automata model to study pedestrian evacuation under different levels of agent aggressiveness. Their method searched for initial conditions that improve evacuation efficiency, and they compared the genetic algorithm results with theoretical analysis and handcrafted solutions. This paper is closely related to our project because it also applies a genetic algorithm directly to an evacuation setting rather than only to parameter calibration. However, their work focuses on optimizing initial evacuation conditions and aggressiveness in a floor field model. In our project, we instead focus on optimizing local movement rules in a simplified agent-based model with high- and low-mobility agents. We also explicitly study whether the evolved rules help both mobility groups and whether they generalize across different crowd scenarios.

Alqahtani, A., & Sherif, A. (2025). "Inclusive crowd evacuation modeling under heterogeneous mobility

constraints." [1]

Alqahtani and Sherif (2025) applied the social force model to study evacuation dynamics of a heterogeneous crowd that contains normal pedestrians alongside disabled ones (wheel chair and visually impaired agents). They demonstrated that disabled agents significantly increase total evacuation time and congestion. It showed the faster-is-slower effect during a single room evacuation. It relates to our project, we want to simulate a simplified hospital evacuation with a heterogeneous mix of high-mobility and low-mobility agents. In the research of Alqahtani and Sherif they did not explore the underlying interaction rules to mitigate the negative effects. Their parameters are based on empirical evidence. We will also model high- and low-mobility agents, but we add a genetic algorithm to search for the local movement rules that improve overall evacuation, without disadvantaging the slower agents.

3 Approach

To tackle this problem within the project constraints, we will use a simplified vectorized 2D agent-based evacuation model as a black-box simulator, wrapped in a genetic algorithm (GA) that optimizes a small vector of local movement parameters. Each evaluation consists of running the simulation and then computing evacuation metrics.

In the agent-based evacuation model, we will simulate a simple 2D room with walls and one or more exits, inspired by a hospital-like layout. The agents will update in discrete time using simple local rules, such as attraction toward the exit, repulsion from nearby agents, and a small amount of noise. This choice is inspired by pedestrian models based on local interaction rules, such as the social force model of Helbing and Molnár [4], which shows how crowd-level behaviour can emerge from simple individual interactions. We will model heterogeneity by including both high-mobility and low-mobility agents, where the main difference between the two groups is their preferred movement speed. This is motivated by the work of Alqahtani and Sherif [1], who show that mobility constraints can significantly affect evacuation time, congestion, and faster-is-slower effects.

The GA will optimize a continuous parameter vector θ representing the weights and thresholds of the local movement rules. Possible parameters include exit attraction strength, repulsion strength, perception radius, and noise level. In contrast to Cui et al. [2], who used differential evolution to calibrate pedestrian model parameters to empirical trajectory data, we will use evolutionary search to optimize evacuation performance directly. This is closer in spirit to the work of Cui et al. [3], who incorporated a GA into an evacuation model to improve evacuation-related outcomes. However, while their work focuses on optimizing initial conditions based on agent aggressiveness, we focus on optimizing local movement rules in a heterogeneous crowd with different mobility levels.

For every candidate parameter vector, the GA will run a fixed number of evacuation simulations to account for randomness in the initial conditions and movement dynamics. The fitness function will combine several criteria, including overall evacuation time, congestion near exits, and collisions or near-collisions. We will also include a penalty for large differences between the evacuation times of the two mobility groups, so that the learned solution does not improve performance only for the high-mobility agents.

We chose this approach because the mapping from local movement rules to evacuation performance is non-linear, noisy, and non-differentiable. GAs are therefore a suitable choice for this black-box, simulation-based optimization problem. They allow us to search over multiple continuous parameters without requiring gradient information, which makes them appropriate for tuning emergent dynamics in a heterogeneous crowd.

We will build both the agent-based simulator and the GA from scratch using standard Python libraries.

We explicitly avoid heavy object-oriented simulation frameworks and external evolutionary optimization libraries, because we aim to keep the model transparent and computationally light. By using a vectorized matrix-based implementation for the kinematic rules, we expect the simulation to be fast enough to evaluate many candidate solutions within the project time frame. The room layouts will be defined directly in code as simple 2D geometric boundaries, so we do not require specialized equipment or external architectural data. To keep the baseline mechanics realistic, we will use empirical findings from Alqahtani and Sherif [1] and Cui et al. [2] as guidance for default movement speeds, spatial radii, and other initial parameter choices.

4 Planning

We will split the project in four phases. First, we implement a minimum viable prototype that consist of a simplified 2D evacuation simulator and a working GA loop (8 - 20 April). Second, we design and run the baseline experiments comparing our GA against random search and fixed heuristic settings (21 April - 13 May). Third, we extend the experiments to group-specific outcomes and generalization across densities, mobility ratios, or layouts (14 - 23 May). Last, we analyse the results, prepare the flash talk, and write the final report (24 May - 12 June).

With those checkpoints we aim to keep the project feasible. By 20 April, we aim to have the prototype and first results. By 13 May, we aim to have the first analysis. By 24 May, we aim to have a clear summary of methods and some early conclusions for the flash talk. By 28 May we aim to have a complete draft for wrapping up the report. If we are not making a deadline we will simplify the model by reducing the amount of parameters, layouts, or repetitions. With this we ensure there is enough time to experiment and analyse.

References

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- [4] Dirk Helbing and Péter Molnár. “Social force model for pedestrian dynamics”. In: *Physical Review E* 51.5 (1995), pp. 4282–4286. DOI: [10.1103/PhysRevE.51.4282](https://doi.org/10.1103/PhysRevE.51.4282).